

Exercise -1 Solutions

1.1 Safety vs Security

Explain the difference between safety and security in the context of machine learning systems. Give one concrete example of a safety concern and one security concern for an ML-based autonomous vehicle.

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Difference Between Safety and Security →

Safety	Security
Deals with accidental failures and unintended harm or to prevent the failure/harm (in case it happens)	Deals with intentional attacks by malicious actors.
The goal is to ensure the system operates reliably - Reliability	Goal is to protect the system from manipulation and unauthorized access - Secure the system
Focuses on robustness and fail-safe behavior - Fault Tolerance	Focuses on defenses such as authentication and attack prevention - Authentication/Privacy

Examples for each case:

Safety Concern for an Autonomous Vehicle → The perception model fails to detect a pedestrian at night, causing the vehicle not to brake and resulting in an accident.

Security Concern (Autonomous Vehicle) → An attacker places adversarial stickers on a traffic sign, causing the model to misclassify it and make an unsafe decision.

1.2 The EU AI Act classifies AI systems into risk categories.

1. Name the main risk categories and briefly describe each.
2. Into which category would you place a perception system for a) an autonomous

vehicle b) a ChatBot? Justify your answer.

3. Research: What obligations does the Act place on providers of systems in that category?

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Main Risk Categories according to the EU act →

1. **Unacceptable Risk** – These AI systems are considered too dangerous because they violate fundamental rights or can cause serious societal harm. Their use is generally banned within the European Union.
e.g. social scoring.
2. **High Risk** – These systems are allowed, but they are subject to extensive legal requirements because failures could significantly affect health, safety, or fundamental rights
eg. autonomous vehicles
3. **Limited Risk** – These systems are considered relatively low risk, but users must be informed that they are interacting with AI or that content was AI-generated.
e.g. chatbots etc.
4. **Minimal Risk** – These are common AI applications with negligible impact on safety or rights. They are largely exempt from specific regulatory requirements.
e.g. spam filters, recommender systems etc.

Classification →

i) Autonomous Vehicle Perception System → **High Risk**, because failures can directly endanger human life.

ii) ChatBot → Usually **Limited Risk**, because users must be informed they are interacting with AI.

(NOTE - Depends on the usage, If it is used in healthcare or employment, it may become **High Risk**)

Obligations for High-Risk Systems (EU AI Act) →

There must be following requirements/processes that need to be in place for high risk models/systems:

- High-risk systems must have a **risk management process** in place throughout the entire lifecycle not just at deployment because risks evolve as the system encounters real-world conditions that weren't anticipated during development.
- **High-quality datasets and documentation** are required because a model is only as trustworthy as the data it learned from. If a system harms someone, regulators and courts need to trace why and that's only possible if training data, design choices, and model versions are properly recorded.
- **Logging and human oversight** must exist because no autonomous system should operate as a black box in safety-critical settings. If an AV perception model makes a wrong call, there must be a record of what inputs it saw and a human who could have intervened. This is what makes accountability legally enforceable.
- **Robustness and accuracy requirements** ensure the system doesn't just work in lab conditions. For an AV, performing well on a sunny test track but failing in rain isn't acceptable. The Act demands evidence that the system holds up across the conditions it will actually face.

1.3 Build the Right Model vs. Build the Model Right

Two complementary perspectives on ML quality (or software quality in general) can be summarized as:

- “Build the right model” — correctness: does the model solve the intended problem?
- “Build the model right” — robustness: is the model built and deployed with adequate care?

For a pedestrian-detection model in an autonomous vehicle, describe one failure captured by each perspective.

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“Build the Right Model” → The model must solve the intended problem. In this case, the model might not give the expected outcome at all due to some functional issues.

Failure Example: The model is trained to detect people standing upright but not children

or disabled/wheelchair users, so it does not correctly solve the pedestrian detection problem. The model must take into account - all forms of pedestrians.

“Build the Model Right” → The model must be engineered, tested, and deployed properly. For instance there can be issues with labelling, training set etc.

Failure Example: The model is trained only on sunny daytime images and is deployed at night without validation, causing failures under unseen conditions. The model’s training set must cover all possible scenarios while training.

1.4 First Impressions

Before looking at any data or model, think about the system described above.

1. Name three situations in which you would not trust this model’s predictions. Explain Why?

2. Which of the three output labels do you consider most safety-critical if missed? Justify your choice.

3. What additional information would you want before deploying this model in a real vehicle?

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Three Situations Where I Would Not Trust the Model →

It is given that - The model is trained only on sunny daytime data and feeds predictions into autopilot logic.

i) During Night → The training data contains only daytime images, so the model may fail when lighting conditions change.

ii) Rain, Fog, or Snow → Weather conditions were not represented during training, leading to distribution shift.

iii) Unusual Road Situations → Construction zones, emergency vehicles, or unusual objects may be outside the training distribution.

Output labels that is most safety-critical if missed →

Pedestrian present is the most critical. Missing a vehicle or traffic light is dangerous, but a missed pedestrian — especially a child stepping onto the road gives the autopilot no chance to brake. Vehicles and lights have more predictable behavior; pedestrians do not. Can cause loss of life.

Additional information I would want before deploying this model in a real vehicle →

Performance metrics broken down by condition (night, rain, edge cases) — not just overall accuracy → Imagine the model is 95% accurate overall, sounds great. But if 90% of training images were sunny daytime shots, that 95% is basically just the model being good at sunny conditions. It might perform at 60% in the rain. Before deploying, the accuracy needs to be sliced by condition at night, fog, rain, construction zones. Overall accuracy can be misleading.

False negative rate specifically for pedestrians → A false negative means the model missed something that was actually there. It predicted "no pedestrian" when a pedestrian existed. This is far more dangerous than a false positive (braking when no one is there). This rate should be as close to zero as possible for pedestrians specifically. Overall error rate doesn't capture this. It needs to be broken out per label.

How the autopilot behaves when a classifier is uncertain or outputs a borderline probability → The model doesn't just output "pedestrian: yes/no" but it outputs a probability, like 0.51 or 0.49. These borderline cases are the most dangerous because the model is essentially saying "I have no idea." A system with no defined behavior for uncertain predictions is a system waiting to fail at exactly the wrong moment.

Whether there's any fallback or human override mechanism → No model is perfect, and edge cases will always exist that weren't in the training data. So, what happens when the model is wrong? Is there a human driver who can take over? Does the car default to a safe stop?

NOTE - Think of these four points as a checklist that moves you from "the model works in the lab" to "the model is safe enough to put on a real road."

1.5 Course Topics and the Safety Case

This seminar covers: testing and validation, explainability, uncertainty estimation, anomaly detection, adversarial machine learning, and alignment. For each topic, briefly explain how it could contribute evidence to the safety case for the CARLA model.

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Topic	Contribution to the Safety Case
Testing and Validation	Shows performance under expected conditions.
Explainability	Verifies that the model relies on relevant image regions.
Uncertainty Estimation	Indicates if the predictions are unreliable.
Anomaly Detection	Detects out-of-distribution inputs.
Adversarial Machine Learning	Assesses robustness against malicious attacks (Security within the safety of the model)
Alignment	Ensures the system behavior supports safety goals